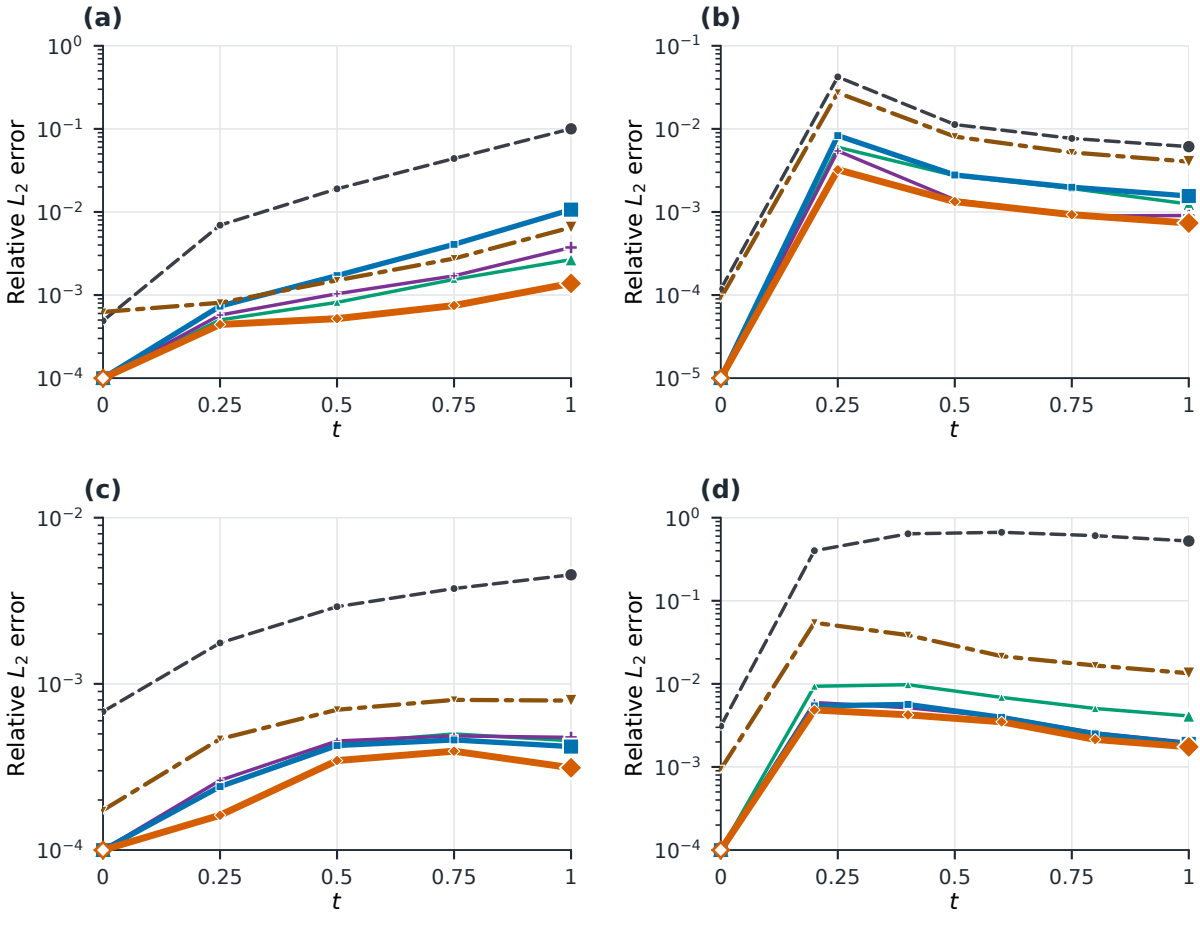
(a)
(b)
(c)
(d)
Relative L_2 error
t
Plain PINN
Time windows (4× steps)
Exact BC + IC
Exact + RBA
Exact + SA-PINN
Exact + causal